

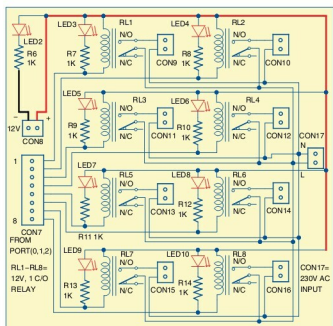


Fig. 4: Circuit of the relay board



Fig. 5: BlueTerm

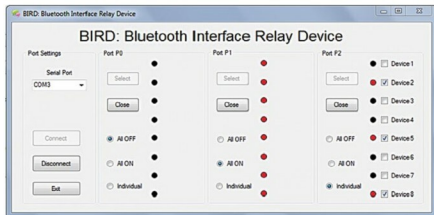


Fig. 6: Screenshot of the Bluetooth application program

at up to 40MHz crystal frequency. It supports 12 clocks per machine cycle (default) and 6 clocks per machine cycle modes, which can be selected via software. The Flash program memory supports both parallel programming and serial programming.

The microcontroller communicates with the Bluetooth module through its TXD (P3.1) and RXD (P3.0) pins. The serial control command sent through the laptop or Android phone is received by the Bluetooth module and processed by microcontroller IC1. The corresponding port pins are enabled as per Table I to switch on various appliances. The microcontroller is programmed in

such a way that one can control individual pins as well as the complete port, simultaneously.

An 11.0592MHz crystal connected between pins 18 and 19, along with two 33pF capacitors, provides the basic clock frequency to microcontroller IC1.

Bluetooth module. A 34-pin, HC-05 Bluetooth module is used in this project. And as shown in Fig. 2 and Fig. 3, only four pins are used. It is an easy-to-use Bluetooth serial port protocol (SPP) module, designed for a transparent wireless serial connection set-up. You can use it simply for a serial port replacement to establish a connection between a PC and your embedded project. The technical specifications of the module are as follows:

1. Bluetooth protocol: Bluetooth specification v2.0+EDR
2. Frequency: 2.4GHz ISM band
3. Modulation: GFSK (Gaussian Frequency Shift Keying)
4. Emission power: $\leq 4\text{dBm}$, Class 2
5. Sensitivity: $\leq -84\text{dBm}$ at 0.1% BER
6. Speed: Asynchronous: 2.1Mbps (max.)/160 kbps, Synchronous: 1Mbps/1Mbps
7. Security: Authentication and encryption
8. Profiles: Bluetooth serial port
9. Power supply: +3.3V DC, 50mA
10. Working temperature: -20° to $+75^{\circ}$ centigrade
11. Dimensions: 26.9mm $\times$ 13mm $\times$ 2.2mm

Relay driver. The ULN2803 is a monolithic, high-voltage and high-current Darlington transistor array. It consists of eight npn Darlington pairs that have high-voltage outputs for switching inductive loads. The collector-current rating of a single Darlington pair is 500mA. Relays are connected to output pins 11 through 18 for each relay driver. The microcontroller drives the relays through ULN2803 as per source program.

Relay board. Fig. 4 shows the circuit of the relay board, where the outputs from the relay driver and loads are terminated. One such board is required for each port (eight appli-